

Inverse Design of Light-Trapping Layers

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Abstract

This report details the surrogate optimization and active learning approaches used to inversely design light-trapping layers.

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1 Introduction

Introduce the problem of light-trapping layers here [1].

2 Methods and theory

2.1 Thin Film Solar Cells

2.2 Rigorous Coupled Wave Analysis

This subsection will introduce the concept of Rigorous Coupled Wave Analysis (RCWA), a method in numerical photonics suited well for stacked periodic structures in the x and y axis with a period Λ_x, Λ_y respectively, and where each layer is homogeneous in the z -direction [4]. The structure's optical properties are completely described by the relative permittivity ϵ and relative permeability μ of each layer, meaning that each structure will have a geometry described by

$$\epsilon(x, y, z)_N = \epsilon(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z} \quad (1)$$

and

$$\mu(x, y, z)_N = \mu(x + m \cdot \Lambda_x, y + n \cdot \Lambda_y)_N, m, n \in \mathbb{Z}, \quad (2)$$

where m and n are integers and the subscript N denotes the N 'th layer of the structure. For the entirety of this project, the designed fourier patterned light trapping layers will be designed using crystalline silicon (Si), Titanium Dioxide (TiO_2) and Silicon Nitride (Si_3N_4), which are all nonmagnetic, so the relative permeability $\mu = 1$. In general, ϵ and μ will be 3×3 tensors, but for isotropic and linear materials these are scalars. The equations solved for are of course the Maxwell equations, namely Faraday's law

$$\nabla \times \mathbf{E} = -\mu_0 \mu(x, y) \frac{\partial \mathbf{H}}{\partial t} \quad (3)$$

and Ampere's law

$$\nabla \times \mathbf{H} = \epsilon_0 \epsilon(x, y) \frac{\partial \mathbf{E}}{\partial t}, \quad (4)$$

Where ϵ_0 and μ_0 are the vacuum permittivity and vacuum permeability respectively. The structures are assumed free of charge and current,

$$\nabla \cdot (\epsilon \mathbf{E}) = 0$$

and

$$\nabla \cdot (\mu \mathbf{H}) = 0.$$

At a single frequency and in homogeneous media, the eigenmodes of Maxwell's equations are solved by plane waves, see [2] section 9.2.1 for example, meaning

$$\mathbf{E} = \mathbf{E}_0 \exp i(\mathbf{k} \cdot \mathbf{r} - \omega t) \quad (5)$$

and

$$\mathbf{H} = \mathbf{H}_0 \exp i(\mathbf{k} \cdot \mathbf{r} - \omega t), \quad (6)$$

Where $\mathbf{E}_0$ and $\mathbf{H}_0$ are the (complex) amplitudes of the fields, with the physical fields being the real parts of $\mathbf{E}$ and $\mathbf{H}$, $\mathbf{k}$ being the wavevector, i.e. the spatial frequency, and ω being the frequency of the fields, also given by the dispersion relation $\omega = \frac{c|\mathbf{k}|}{n(\lambda)}$, where λ is the wavelength of the light. It is noted that conservation of momentum leads to the spatial components of the wavevector only having two degrees of freedom, i.e.

$$|\mathbf{k}| = \frac{2\pi n(\lambda)}{\lambda} = \sqrt{k_x^2 + k_y^2 + k_z^2} = k_0, \quad (7)$$

and in general, for an incident angle θ and azimuthal angle ϕ , the x and y components will be chosen by

$$(k_x, k_y) = k_0(\sin \theta \cos \phi, \sin \theta \sin \phi). \quad (8)$$

Inserting equations 5 and 6 in equations 3 and 4 converts the time derivative to an algebraic factor of $-i\omega t$, meaning that the equations appear in their time harmonic form, common to many frequency domain numerical methods in photonics,

$$\nabla \times \mathbf{E} = i\omega\mu_0\mu(x, y)\mathbf{H} \quad (9)$$

$$\nabla \times \mathbf{H} = -i\omega\epsilon_0\epsilon(x, y)\mathbf{E}. \quad (10)$$

Owing to the periodicity of the system, see equations 1 and 2, both the electromagnetic fields and the permittivity and permeability may be written as their truncated fourier series in the x- and y-directions as below. Truncating the electromagnetic fields to order M, N in x, y makes RCWA a semi-analytical method precise only to this fourier order. The final fields are reconstructed as an inverse Fast Fourier Transform, so including enough fourier orders is an important convergence criterion.

$$\epsilon(x, y) = \sum_{m=-2M}^{2M} \sum_{n=-2N}^{2N} \epsilon_{m,n} \exp i(mL_x x + nL_y y), \quad (11)$$

$$\mu(x, y) = \sum_{m=-2M}^{2M} \sum_{n=-2N}^{2N} \mu_{m,n} \exp i(mL_x x + nL_y y), \quad (12)$$

$$\mathbf{E} = \exp i\mathbf{k} \cdot \mathbf{r} \sum_{m=-M}^M \sum_{n=-N}^N \mathbf{E}_{m,n} \exp i(mL_x x + nL_y y), \quad (13)$$

$$\mathbf{H} = \exp i\mathbf{k} \cdot \mathbf{r} \sum_{m=-M}^M \sum_{n=-N}^N \mathbf{H}_{m,n} \exp i(mL_x x + nL_y y), \quad (14)$$

where $L_x = \frac{2\pi}{\Lambda_x}$ and $L_y = \frac{2\pi}{\Lambda_y}$, and the parameters $\epsilon_{m,n}, \mu_{m,n}, \mathbf{E}_{m,n}$ and $\mathbf{H}_{m,n}$ are the fourier amplitudes of the m, n 'th diffraction order. It is worth mentioning, that since the momentum of light is always conserved, k_z implicitly depends on the expansion orders m, n , and writing out the k-vector in full, including the fourier components, yields

$$\mathbf{k}_{m,n} = (k_x + mL_x, k_y + nL_y, k_z(m, n)) = \quad (15)$$

$$= \left(k_x + mL_x, k_y + nL_y, \sqrt{k_0^2 - (k_x + mL_x)^2 - (k_y + nL_y)^2} \right), \quad (16)$$

where equation 7 has been used. The physical interpretation of the fourier series of the electromagnetic fields is then made obvious; the different terms are different diffraction orders, each with part of the initial z-component momentum kicked into either the x- or y-direction. It is noted how the permittivity and permeability is expanded to twice the order of the electromagnetic fields E and H . This arises from the fact that the algebraic components of equations 3 and 4 include products of Fourier series, so to reach order M, N in the electromagnetic fields, the permittivity and permeability must be truncated at twice the order. An example for just one fourier order will be given below, but for more details, I refer to [3]. By inserting

2.3 Finite Element Method

2.4 Machine Learning Surrogates

2.5 Optimization Strategies

3 Results

Present the performance of the surrogate models and the final inverse design outcomes.

Figure 1: Placeholder for a results figure.

4 Discussion

Discuss the results and what worked/didn't work.

5 Outlook

Discuss future work.

6 Conclusion

Summarize the findings.

References

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